

Lattice-Free Open Vocabulary Keyword Spotting

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Introduction

- Task of searching words or group of words embedded in unconstrained speech.
- Along with the keyword's presence, pinpointing the location is vital.
- Keyword Spotting systems rely on predefined keywords.
- Typical statistical methods use phoneme recognizers to localize the keyword¹.

¹R. C. Rose and D. B. Paul, A hidden markov model based keyword recognition system

Motivation

- Extracting the phone lattices and searching the keywords is computationally expensive.
- These approaches are limited to number of keywords.
- Out-of-vocabulary keywords ?
- Do we really need phoneme recognizers to extract phone lattices?
- Given a keyword $q = (q_1, q_2, \dots, q_K)$ and acoustic features $X = (x_1, x_2, \dots, x_N)$, seek for $y = (y_1, y_2, \dots, y_N)$

$$y = \left\{ \begin{array}{ll} 1, & \text{if } q \text{ is spoken in } X \text{ in a time span} \\ 0, & \text{otherwise} \end{array} \right\} \quad (1)$$

- Can Neural networks solve?

- Projecting the audio and keyword into common latent space and making the decision
- ASR-free KWS system² predicts the presence of the keyword within the given audio.
- End-to-end open-vocabulary keyword spotting³ method predicts keyword endpoints using CTC.
- We introduce a lattice-free KWS method that accurately detects and locates the in-vocabulary and out-of-vocabulary keywords.

²Audhkhasi, et al. End-to-end ASR-free keyword search from speech. ICASSP (2017).

³Wei, Bo et al. End-to-End Transformer-Based Open-Vocabulary Keyword Spotting with Location-Guided Local Attention. Interspeech (2021).

Architecture

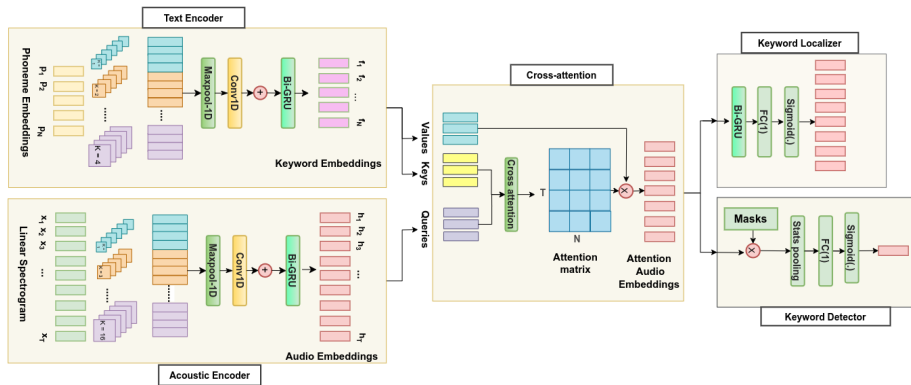


Figure: Architecture of the proposed method.

t-SNE visualisation of the phoneme embeddings

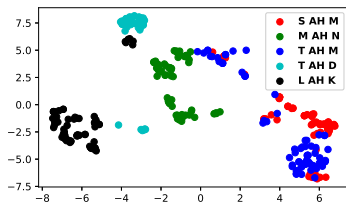


Figure: t-SNE visualisation of the /AH/ phoneme occurring in different context

- Text encoder learns to discriminate the phonemes occurring in different contexts, i.e., /S AH M/, /T AH M/, /M AH N/, /T AH D/, and /L AH K/.

Cross attention

- Cross-attention allows the model to focus on the relevant acoustic features where the text keyword matches the audio.

$$\mathbf{A} = \text{softmax}\left(\frac{\mathbf{QK}^T}{\sqrt{d_k}}\right) \quad (2)$$

$$\mathbf{W} = \mathbf{AV} \quad (3)$$

- where $\mathbf{K}$ and $\mathbf{V}$ are extracted from Text embeddings and $\mathbf{Q}$ are extracted from Acoustic embeddings.
- Attention matrix exhibits monotonic pattern if keyword exists otherwise it exhibits obscure pattern.

Visualisations of boundaries and Attention matrices

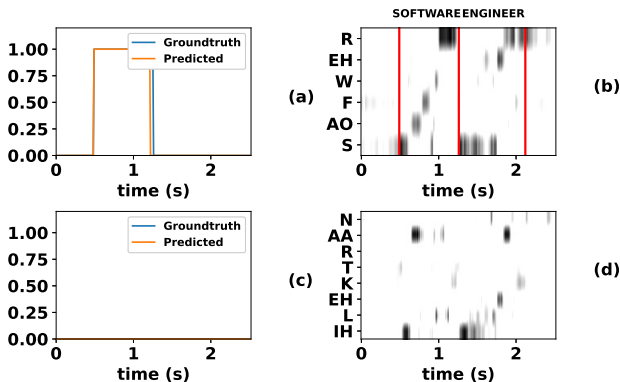


Figure: Visualisations of (a) Ground truth and predicted boundaries for a positive example. (b) Attention matrix for positive example (c) Ground truth and predicted boundaries for a negative example. (d) Attention matrix for negative example

Loss functions

- Binary cross entropy losses employed at the frame level and utterance level are given by

$$L_{loc} = -\frac{1}{V} \sum_{i=1}^V \sum_{t=1}^T (y_t^i \log(p_t^i) + (1 - y_t^i) \log(1 - p_t^i))$$

$$L_{det} = -\frac{1}{V} \sum_{i=1}^V (y^i \log(p^i) + (1 - y^i) \log(1 - p^i))$$

$$L_{total} = \lambda_1 L_{loc} + \lambda_2 L_{det}$$

$y_t^i \rightarrow$ ground truth posterior probability

$p_t^i \rightarrow$ predicted posterior probability

Dataset description

- The proposed method is evaluated on an Indian accented English speech data set collected from IITM ASR Challenge⁴.
- The proposed method is evaluated in 2 scenarios, i.e., in-vocabulary and out-of-vocabulary.
- In-vocabulary scenario : keywords seen, audios unseen
- Out-of-vocabulary scenario : keywords unseen, audios unseen
- Out-of-vocabulary speech corpus is collected from NPTEL site⁵

⁴<https://sites.google.com/view/englishasrchallenge/>

⁵<https://nptel.ac.in/courses>

Training details

- Randomly select keywords from sentences containing more than 5 phonemes to create positive examples.
- Randomly select keywords from the lexicon that do not appear in the given utterance to generate negative examples.
- 128-dimensional linear spectrogram is extracted at a frame shift of 10ms and window size of 25ms.
- The model is trained for 100 epochs with an ADAM optimizer with a learning rate of $1e-3$.
- The learning rate is reduced by 0.1 with patience of 30 epochs.

Evaluation in in-vocabulary scenario

- The performance of the 2 models is evaluated for 10 in-vocabulary keywords, i.e., THOUSAND, INDIA, DECEMBER, CENTURY, FACEBOOK, MINISTER, DIRECTOR, CAPTAIN, TECHNOLOGY, BUSINESS.

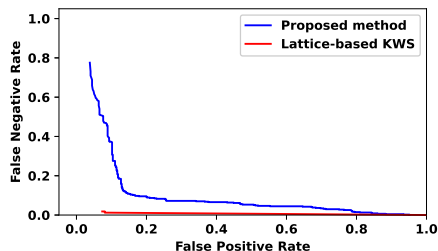


Table: Performance comparison of the 2 models in terms of MDR, FAR, and EER

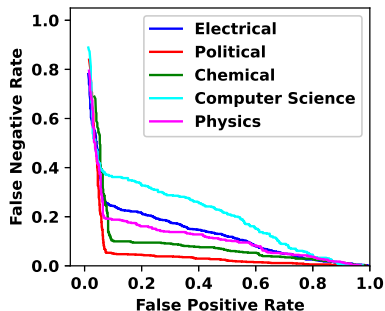
System	MDR	FAR	EER
Proposed method	8.26	24.15	13.98
Lattice-based KWS	1.41	8.08	6.98

Evaluation in out-of-vocabulary scenario

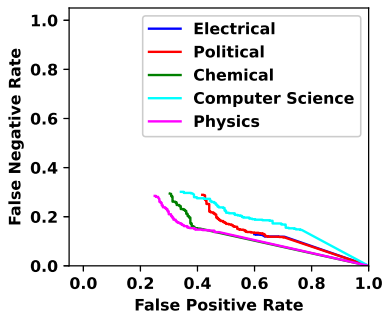
Table: Keywords chosen for out-of-vocabulary scenario

Domain	Keywords	Amount
Electrical	INFINITY,DISTORTIONLESS, CONVOLUTION, MAGNITUDE, INTEGRAL, SUMMATION, CONVERGENCE, LAPLACE, IMAGINARY, TRANSITION	7.5h
Chemical	OXIDATION, AMMONIA, COPPER, GEOMETRY, COBALT, CHLORIDE, MOLECULE, POTASSIUM, DISTILLATION, NITRATE	7.5h
Computer Science	PYTHON, EXECUTE, PROGRAMMING, REMAINDER, INTERPRETER, DEFINITION, INSTRUCTION, FACTORIAL, RECURSIVE, INDENTATION	5.5h
Physics	VECTOR, TRAJECTORY, PARTICLE, EQUILIBRIUM, KINETIC, DIVERGENCE, VELOCITY, OSCILLATOR, LAGRANGIAN, AMPLITUDE	7h
Political	CITIZENSHIP, DEMOCRACY, ARISTOTLE,CONCEPTION, INTERPRETATION, ETHICS, DOMINANT, CORRUPTING, PHILOSOPHER, MORAL	8h

Evaluation in out-of-vocabulary scenario



(a)



(b)

Figure: Visualisations of DET curves of (a) Proposed model (b) Lattice-based KWS

Evaluation in out-of-vocabulary scenario

Table: Equal Error Rates of proposed method and baseline method

Domain	Proposed method	Lattice-based KWS
Electrical	20.96	40.78
Political	6.67	36.99
Chemical	9.96	30.03
Computer Science	29.19	32.85
Physics	17.63	26.63

- Proposed model consistently outperforms the baseline model across all five domains.
- The proposed system is robust to unseen speakers.
- Model exhibits proficiency in identifying repetitive keywords.

Conclusions & Future work

- Proposed lattice-free open vocabulary keyword spotter capable of localizing the keywords and predicting their existence.
- The results suggest that this approach has robust generalization capabilities in terms of domain mismatch and speaker variability.
- This approach does not require lattice extraction and searching the keyword in the lattice.
- Explore the system development of language-agnostic Keyword Spotting.
- Extending our investigations towards the audio queries.

Thank you